LAB 8

Q1.

When $X = 1$, the operation $R1 \leftarrow R2$ is perfomed.

When $X = 0$ and $Y = 1$, the operation $R1 \leftarrow R3 + R4$

So, RTL statements are:

$X: R1 \leftarrow R2$

$X'Y: R1 \leftarrow R3 + R4$

Incorrect statements:

$X: R2 \leftarrow \sim R2, R1 \leftarrow R0$, two micro-operations are specified under the same control signal

X. This may cause a conflict if the hardware does not support parallel execution of operations.

Y: $R0 \leftarrow R1, R0 \leftarrow R2$, the same register R0 is assigned two different values at the same time, which leads to ambiguity in the final value.

R2: $R2 \leftarrow R2 + 1$, the control function is incorrectly written as a register name. A control function must be a condition or control signal, not a register

Z: R3 ← R4, R3 ← R2 + ~R4+1
two different values are assigned to the same register R3 simultaneously, which results in a conflict.

Q2.

Initial values:

AR = 11110010
BR = 11111111
CR = 10111001
DR = 11101010

AR ← AR + BR

AR(242) + BR(255) = 497

$497 = 111110001_2$

lowes 8 bits: $1111\ 0001_2$

Drop the carry $AR = 1111\ 0001_2$

New $AR(1111\ 0001) = 241$

$CR \leftarrow CR.DR$

CR	DR	CR^DR
1	1	1
0	1	0
1	1	1
1	0	0
1	1	1
0	0	0
0	1	0
1	0	0

New $CR = 1010\ 1000_2 = 168$

$AR \leftarrow AR - CR$

Sub. 2's comp. $AR + (\sim CR + 1)$

2's comp:

$$\begin{array}{r} 01010111 \\ +\quad\quad 1 \\ \hline 01011000 \end{array}$$

AR + 2's compliment

$$\begin{array}{r} 1111\ 0001 \\ 0101\ 1000 \\ \hline 10100\ 1001 \end{array}$$

lowest 8 bit 0100 1001

Overall:

AR = 0100 1001

BR = 1111 1111

CR = 1010 1000

DR = 1110 1010

Q3.

Part A:

For destination register R0:
the sources are R1 and R2

For destination register R1:
the source is R4

For destination register R2:

the sources are R3 and R4

For destination register R3:
the source is R1

For destination register R4:
the sources are R0 and R2

Part B

For source register R0, the destination is R4
For source register R1, the destinations are R0 and R3
For source register R2, the destinations are R0 and R4
For source register R3, the destination is R2
For source register R4, the destinations are R1 and R2

Q4

T3: $\{D_0, \ldots, D_7\} = DEC(IR(12-14))$,
$AR = IR(0-11)$, $I = IR(15)$

1) $\{D_0, \ldots, D_7\} = DEC(IR(12-14))$
- bits R-14 of instructor reg. are used as input to a 3-8 decoder
- decoder activates exactly one of eight output lines
- this enables and identifies the control circuit

2) $AR \leftarrow IR(0-1)$
- 12 least sign. bits are transfered to address register
- this sets up memory address for subsequent operant fetch or store

3) $I \leftarrow IR(15)$

- the most significant bit is transfered to the indirect flip-flop
- bit determines addressing mode for direct address.

Q5

The common fetch cycle consists of following steps:

At time T0, the content of the program counter are transferred to the address register ($AR \leftarrow PC$)

At time T1, the instruction is fetched from memory into the data register and the

counter is incremented ($DR \leftarrow M[AR]$, $PC \leftarrow PC+1$)

At time T2: the instruction is loaded into the instruction register, the address field is transferred to the address register, and the mode bit is stored ($IR \leftarrow DR$, $AR \leftarrow (IR(0,11))$, $I \leftarrow IR(15)$)

In direct addressing ($I=0$), at time T3 the operand is fetched directly from the memory location specified in the address register ($DR \leftarrow M[AR]$)

In indirect addressing ($I=1$) at time T3 the effective address is obtained from memory and loaded into the address register ($AR \leftarrow M[AR]$). At time T4, the operand is fetched ($DR \leftarrow M[AR]$)